CHAPTER 9

Profit and Loss

INTRODUCTION

Nowadays, transactions have become a common feature of life. When a person deals in the purchase and sale of any item, he either gains or loses some amount generally. The aim of any business is to earn profit. The commonly used terms in dealing with questions involving sale and purchase are:

Cost Price: The cost price of an article is the price at which an article has been purchased. It is abbreviated as C.P.

Selling Price: The selling price of an article is the price at which an article has been sold. It is abbreviated as S.P.

Profit or Gain: If the selling price of an article is more than the cost price, there is a gain or profit.

Thus, Profit or Gain = S.P. − C.P.

Loss: If the cost price of an article is greater than the selling price, the seller suffers a loss.

Thus, Loss = C.P. − S.P.

Note that profit and loss are always calculated with respect to the cost price of the item.

Illustration 1: (i) If C.P. = ₹235, S.P. = ₹240, then profit = ?

(ii) If C.P. = ₹116, S.P. = ₹107, then loss = ?

Solution: (i) Profit = S.P. − C.P. = 240 − 235 = ₹5.

(ii) Loss = C.P. − S.P. = 116 − 107 = ₹9.

BASIC FORMULAE

01 Gain on ₹100 is *Gain per cent*

$$\text{Gain \%} = \frac{\text{Gain} \times 100}{\text{C.P.}}$$

Loss on ₹100 is *Loss per cent*

$$\text{Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

Illustration 2: The cost price of a shirt is ₹200 and selling price is ₹250. Calculate the % of profit.

Solution: We have, C.P. = ₹200, S.P. = ₹250.

Profit = S.P. − C.P. = 250 − 200 = ₹50.

$$\therefore \quad \text{Profit \%} = \frac{\text{Profit} \times 100}{\text{C.P.}} = \frac{50 \times 100}{200} = 25\%$$

Illustration 3: Anu bought a necklace for ₹750 and sold it for ₹675. Find her percentage of loss.

Solution: Here, C.P. = ₹750, S.P. = ₹675.

Loss = C.P. − S.P. = 750 − 675 = ₹75.

$$\therefore \quad \text{Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{75 \times 100}{750} = 10\%$$

02 When the selling price and gain % are given:

$$\text{C.P.} = \left(\frac{100}{100 + \text{Gain\%}}\right) \times \text{S.P.}$$

03 When the cost and gain per cent are given:

$$\text{S.P.} = \left(\frac{100 + \text{Gain\%}}{100}\right) \times \text{C.P.}$$

Explanation

$$\text{Since Profit \%} = \frac{\text{Profit} \times 100}{\text{C.P.}}$$

$$= \left[\frac{(\text{S.P.} - \text{C.P.}) \times 100}{\text{C.P.}}\right]$$

$$\therefore \quad \frac{\text{Profit \%}}{100} = \frac{\text{S.P.}}{\text{C.P.}} - 1$$

$$\text{or,} \quad \frac{\text{S.P.}}{\text{C.P.}} = 1 + \frac{\text{Profit \%}}{100}$$

$$\therefore \quad \text{S.P.} = \left(\frac{100 + \text{Profit \%}}{100}\right) \times \text{C.P.}$$

$$\text{and, C.P.} = \left(\frac{100}{100 + \text{Profit \%}}\right) \times \text{S.P.}$$

04 When the cost and loss per cent are given:

$$\text{S.P.} = \left(\frac{100 - \text{Loss \%}}{100}\right) \times \text{C.P.}$$

05 When the selling price and loss per cent are given:

$$\text{C.P.} = \left(\frac{100}{100 - \text{Loss \%}}\right) \times \text{S.P.}$$

Explanation

$$\text{Since Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

$$= \left[\frac{(\text{C.P.} - \text{S.P.}) \times 100}{\text{C.P.}}\right]$$

$$\therefore \quad \frac{\text{Loss\%}}{100} = 1 - \frac{\text{S.P.}}{\text{C.P.}}$$

$$\text{or} \quad \frac{\text{S.P.}}{\text{C.P.}} = 1 - \frac{\text{Loss\%}}{100}$$

$$\therefore \quad \text{S.P.} = \left(\frac{100 - \text{Loss\%}}{100}\right) \times \text{C.P.}$$

$$\text{and,} \quad \text{C.P.} = \left(\frac{100}{100 - \text{Loss\%}}\right) \times \text{S.P.}$$

Illustration 4: Mr Sharma buys a cooler for ₹4500. For how much should he sell it to gain 8%?

Solution: We have, C.P. = ₹4500, gain % = 8%

$$\therefore \quad \text{S.P.} = \left(\frac{100 + \text{Gain \%}}{100}\right) \times \text{C.P.}$$

$$= \left(\frac{100 + 8}{100}\right) \times 4500$$

$$= \frac{108}{100} \times 4500 = ₹4860.$$

Illustration 5: By selling a fridge for ₹7200, Pankaj loses 10%. Find the cost price of the fridge.

Solution: We have, S.P. = ₹7200, loss % = 10%

$$\therefore \quad \text{C.P.} = \left(\frac{100}{100 - \text{Loss\%}}\right) \times \text{S.P.}$$

$$= \left(\frac{100}{100 - 10}\right) \times 7200$$

$$= \frac{100}{90} \times 7200 = ₹8000.$$

Illustration 6: By selling a pen for ₹99, Mohan gains $12\frac{1}{2}$%. Find out cost price of the pen.

Solution: Here, S.P. = ₹99, gain % = $12\frac{1}{2}$% or $\frac{25}{2}$%

$$\therefore \quad \text{C.P.} = \left(\frac{100}{100 + \text{Gain\%}}\right) \times \text{S.P.}$$

$$= \left(\frac{100}{100 + \frac{25}{2}}\right) \times 99$$

$$= \left(\frac{100 \times 2}{225}\right) \times 99 = ₹88.$$

SOME USEFUL SHORTCUT METHODS

01 If a man buys x items for ₹y and sells z items for ₹w, then the gain or loss per cent made by him is

$$\left(\frac{xw}{zy} - 1\right) \times 100\%$$

Explanation

S.P. of z iterms = ₹w

S.P. of x items = ₹$\frac{w}{z}x$

Net profit = $\frac{w}{z}x - y$.

$$\therefore \quad \% \text{ Profit} = \frac{\frac{w}{z}x - y}{y} \times 100\%$$

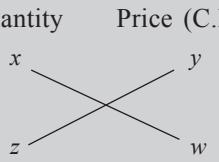
i.e., $\left(\frac{xw}{zy} - 1\right) \times 100\%$

which represents loss, if the result is negative.

Notes

In the case of gain per cent, the result obtained bears positive sign whereas in the case of loss per cent the result obtained bears sign negative.

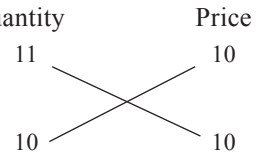
How to Remember?



1. Cross-multiply the numbers connected by the arrows (xw and zy).
2. Mark the directon of the arrows for cross-multiplicaton. The arrow going down forms the numerator while the arrow going up forms the denominator $\left(\frac{xw}{zy}\right)$.

Illustration 7: If 11 oranges are bought for ₹10 and sold at 10 for ₹11, what is the gain or loss %?

Solution:



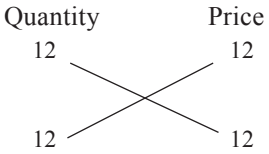
$$\% \text{ profit} = \left(\frac{xw}{zy} - 1\right) \times 100\%$$

$$= \left(\frac{11 \text{ x } 11}{10 \text{ x } 10} - 1\right) \times 100\%$$

$$= \frac{21}{100} \times 100\% = 21\%$$

Illustration 8: A fruit seller buys apples at the rate of ₹12 per dozen and sells them at the rate of 15 for ₹12. Find out his percentage gain or loss.

Solution:



$$\% \text{ gain or loss} = \left(\frac{xw}{zy} - 1\right) \times 100\%$$

$$= \left(\frac{12 \times 12}{15 \times 12} - 1\right) \times 100\%$$

$$\left(\frac{144}{180} - 1\right) \times 100\%$$

$$\left(\frac{-36}{180} \times 100\%\right)$$

Since the sign is –ve, there is a loss of 25%

02 If the cost price of m articles is equal to the selling price of n articles, then

$$\% \text{ gain or loss} = \left(\frac{m-n}{n}\right) \times 100$$

[If $m > n$, it is % gain and, if $m < n$, it is % loss]

Explanation

Let, the C.P. of an article be ₹1.

$\therefore$ C.P. of m articles = ₹m × 1 = ₹m

$\because$ S.P. of n articles = ₹m

$\therefore$ S.P. of an article = ₹$\frac{m}{n}$

$\therefore$ Profit on 1 article = ₹$\left(\frac{m}{n} - 1\right)$

i.e., ₹$\left(\frac{m-n}{n}\right)$

$\therefore$ % profit = $\left(\frac{m-n}{n}\right) \times 100$ i.e., $\left(\frac{m-n}{n}\right) \times 100$

Illustration 9: A shopkeeper professes to sell his goods on cost price, but uses 800 gm, instead of 1 Kg. What is his gain %?

Solution: Here, cost price of 1000 gm is equal to selling price of 800 gm,

$$\therefore \quad \% \text{ gain} = \left(\frac{m-n}{n}\right) \times 100$$

$$= \left(\frac{1000 - 800}{800}\right) \times 100$$

$$= \frac{200}{800} \times 100 = 25\%$$

Illustration 10: If the selling price of 12 articles is equal to the cost price of 18 articles, what is the profit %?

Solution: Here, $m = 18$, $n = 12$

$$\therefore \quad \text{Profit \%} = \left(\frac{m-n}{n}\right)\times 100$$

$$= \left(\frac{18-12}{12}\right)\times 100$$

$$= \frac{6}{12}\times 100 = 50\%$$

03 If an article is sold at a price $S.P._1$., then % gain or % loss is x and if it is sold at a price $S.P_2$., then % gain or % loss is y. If the cost price of the article is C.P., then

$$\frac{S.P._1}{100+x} = \frac{S.P._2}{100+y} = \frac{C.P.}{100} = \frac{S.P._1 - S.P._2}{x-y},$$

where x or y is –ve, if it indicates a loss, otherwise it is +ve.

Illustration 11: By selling a radio for ₹1536, Suresh lost 20%. What per cent shall he gain or lose by selling it for ₹2000?

Solution: Here, $S.P._1 = 1536$, $x = -20$

(–ve sign indicates loss)

$S.P._2 = ₹2000$, $y = ?$

Using the formula:

$$\frac{S.P._1}{100+x} = \frac{S.P._2}{100+y},$$

we get, $\frac{1536}{100-20} = \frac{2000}{100+y}$

$$\Rightarrow \quad 100 + y = \frac{2000\times 80}{1536} = 104\frac{1}{6}$$

$$\Rightarrow \quad y = 4\frac{1}{6}\%$$

Thus, Suresh has a gain of $4\frac{1}{6}\%$ by selling it for ₹2000.

04 If 'A' sells an article to 'B' at a gain/loss of $m\%$ and 'B' sells it to 'C' at a gain/loss of $n\%$. If 'C' pays ₹z for it to 'B', then the cost price for 'A' is

$$\left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

where m or n is –ve, of it indicates a loss, otherwise it is +ve.

Illustration 12: Mohit sells a bicycle to Rohit at a gain of 10% and Rohit again sells it to Jyoti at a profit of 5%. If Jyoti pays ₹462 to Rohit, what is the cost price of the bicycle for Mohit?

Solution: Here, $m = 10$, $n = 5$, $z = ₹462$.

Using the formula,

$$\text{C.P.} = \left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

we get, C.P. for Mohit $= \left[\frac{100^2\times 462}{(100+10)(100+5)}\right]$

$$= \frac{462\times 10000}{110\times 105} = ₹400.$$

Illustration 13: 'A' sells a DVD to 'B' at a gain of 17% and 'B' again sells it to 'C' at a loss of 25%. If 'C' pays ₹1053 to 'B', what is the cost price of the DVD to 'A'?

Solution: We have, $m = 17$, $n = -25$, $z = ₹1053$.

$\therefore$ Cost price of DVD to

$$= \left[\frac{100^2 z}{(100+m)(100+n)}\right]$$

$$= \frac{100\times 100\times 1053}{(100+17)(100-25)}$$

$$= \frac{100\times 100\times 1053}{117\times 75} = ₹1200.$$

05 If 'A' sells an article to 'B' at a gain/loss of $m\%$, and 'B' sells it to 'C' at a gain/loss of $n\%$, then the resultant profit/loss per cent is given by

$$\left(m+n+\frac{mn}{100}\right) \qquad \cdots(1)$$

Where m or n is –ve, if it indicates a loss, otherwise it is +ve.

Notes

The expression given by eq. (1) represents resultant profit or loss according as it is +ve or –ve.

Illustration 14: 'A' sells a horse to 'B' at a profit of 5% and 'B' sells it: to 'C' at a profit of 10%. Find out the resultant profit per cent.

Solution: We have, $m = 5$ and $n = 10$.

$$\therefore \quad \text{Resultant profit \%} = \left(m+n+\frac{mn}{100}\right)$$

$$= \left(5+10+\frac{5\times 10}{100}\right)$$

$$= \frac{31}{2}\% \quad \text{or,} \quad 15\frac{1}{2}\%$$

Illustration 15: Manoj sells a shirt to Yogesh at a profit of 15%, and Yogesh sells it to Suresh at a loss of 10%. Find the resultant profit or loss.

Solution: Here, $m = 15$, $n = -10$

$$\therefore \quad \text{Resultant profit/loss \%} = \left(m + n + \frac{mn}{100}\right)$$

$$= \left(15 - 10 + \frac{15 \times -10}{100}\right) = \left(15 - 10 - \frac{150}{100}\right)$$

$$= 7/2\% \quad \text{or} \quad 3\frac{1}{2}\%,$$

which represents profit as the sign is +ve.

06 When two different articles are sold at the same selling price, getting gain/loss of $x\%$ on the first and gain/loss of $y\%$ on the second, then the overall % gain or % loss in the transaction is given by

$$\left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%$$

The above expression represent overall gain or loss according as its sign is +ve or –ve.

07 When two different articles are sold at the same selling price getting a gain of $x\%$ on the first and loss of $x\%$ on the second, then the overall % loss in the transaction is given by

$$\left(\frac{x}{10}\right)^2 \%$$

Note that in such questions, there is always a loss.

Explanation

Let, each article be sold at ₹z.

Since gain/loss of $x\%$ is made on the first, cost price of the first article

$$= ₹z\left(\frac{100}{100 + x}\right)$$

Also, gain/loss of $y\%$ is made on the second. Therefore, cost price of the second article

$$= ₹z\left(\frac{100}{100 + y}\right)$$

$$\therefore \quad \text{Total C.P.} = z\left(\frac{100}{100 + x}\right) + z\left(\frac{100}{100 + y}\right)$$

$$= z\left[\frac{100(100 + y) + 100(100 + x)}{(100 + x)(100 + y)}\right]$$

Total S.P. $= 2z$.

$$\therefore \quad \text{Overall \% gain or loss} = \frac{\text{S.P.} - \text{C.P.}}{\text{C.P.}} \times 100$$

$$= \frac{2z - \dfrac{100z[100 + x + 100 + y]}{(100 + x)(100 + y)}}{\dfrac{100z[100 + x + 100 + y]}{(100 + x)(100 + y)}} \times 100$$

$$= \frac{2(100 + x)(100 + y) - 100(200 + x + y)}{100(200 + x + y)} \times 100$$

$$= \frac{100x + 100y + 2xy}{(100 + x) + (100 + y)}\%$$

$$= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%.$$

Notes

In case $y = -x$, we have

Overall % gain or loss $= -\dfrac{x^2}{100}\%$,

Since the sign is –ve, there is always a loss.

Illustration 16: Mahesh sold two scooters, each for ₹24000. If he makes 20% profit on the first and 15% loss on the second, what is his gain or loss per cent in the transactions?

Solution: Here, $x = 20$ and $y = -15$.

$$\therefore \quad \text{Over all gain/loss \%}$$

$$= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)}\right]\%$$

$$= \left[\frac{100(20 - 15) + 2 \times 20 \times -15}{(100 + 20) + (100 - 15)}\right]\%$$

$$= -\frac{100}{205}\% = -\frac{20}{41}\%$$

which represents loss, being a –ve expression.

Illustration 17: Rajesh sold two horses for ₹990 each; gaining 10% on the one and losing 10% on the other. Find out his total gain or loss per cent.

Solution: Here, $x = 10$.

$$\therefore \quad \text{Overall loss \%} = \left(\frac{x}{10}\right)^2 \% = \left(\frac{10}{10}\right)^2 \% = 1\%$$

08 A merchant uses faulty measure and sells his goods at gain/loss of $x\%$. The overall % gain/loss(g) is given by

$$\frac{100 + g}{100 + x} = \frac{\text{True measure}}{\text{Faulty measure}}$$

Notes

If the merchant sells his goods at cost price, then $x = 0$.

09 A merchant uses $y\%$ less weight/length and sells his goods at gain/loss of $x\%$. The overall % gain/loss is given by

$$\left[\left(\frac{y+x}{100-y}\right)\times 100\right]\%$$

Illustration 18: A dishonest shopkeeper professes to sell cloth at the cost price, but he uses faulty meter rod. His meter rod measures 95 cm only. Find his gain per cent.

Solution: Here, true measure = 100 cm.

False measure = 95 cm.

Since the shopkeeper sells the cloth at cost price, $\therefore x = 0$.

$\therefore$ Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{\text{True measure}}{\text{Faulty measure}}$$

$$\Rightarrow \quad \frac{100+g}{100} = \frac{100}{95} \Rightarrow 100 + g = \frac{100\times 100}{95}$$

$$\Rightarrow \quad g = \frac{10000}{95} - 100 = 5\frac{5}{19}\%$$

Illustration 19: A dishonest shopkeeper professes to sell his goods at cost price, but he uses a weight of 800 g for the Kg weight. Find out his gain per cent.

Solution: True measure = 1000 g.

False measure = 800 g. Also, $x = 0$.

$\therefore$ Overall gain % is given by

$$\frac{100+g}{100+x} = \frac{\text{True measure}}{\text{False measure}}$$

$$\Rightarrow \quad \frac{100+g}{100} = \frac{1000}{800} \Rightarrow 100 + g = \frac{1000\times 100}{800}$$

$$\Rightarrow \quad g = \frac{1000}{8} - 100 = 25\%$$

Illustration 20: A shopkeeper sells goods at 44% loss on cost price, but uses 30% less weight. What is his percentage profit or loss?

Solution: Here, $x = -44$ and $y = 30$.

$$\therefore \text{ Overall gain/loss\%} = \left(\frac{y+x}{100-y}\right)\times 100\%$$

$$= \left(\frac{30-44}{100-30}\times 100\right)\%$$

$$= \left(\frac{-14}{70}\times 100\right)\% = -20\%,$$

which represents loss being a negative expression.

10 A person buys two items for ₹A and sells one at a loss of $l\%$ and the other at a gain of $g\%$. If each item was sold at the same price, then

(a) The cost price of the item sold at loss

$$= \frac{A(100 + \%\text{gain})}{(100 - \%\text{loss}) + (100 + \%\,\text{gain})}$$

(b) The cost price of the item sold at gain

$$= \frac{A(100 - \%\,\text{loss})}{(100 - \%\text{loss}) + (100 + \%\,\text{gain})}$$

Illustration 21: Ramesh buys two books for ₹410. He sells one at a loss of 20% and the other at a gain of 25%. If both the books are sold at the same price, find out the cost price of two books.

Solution: Cost price of the book sold at a loss of 20%

$$= \frac{410\,(100+25)}{(100-20)+(100+25)}$$

$$= \frac{410\times 125}{80+125} = ₹250.$$

Cost price of the book sold at a profit of 25%

$$= \frac{410\,(100-20)}{(100-20)+(100+25)} = \frac{410\times 80}{80+125}$$

$$= ₹160.$$

11 If two successive discounts on an article are $m\%$ and $n\%$ respectively, then a single discount equivalent to the two successive discounts will be:

$$\left(m + n - \frac{mn}{100}\right)\%$$

Explanation

Let, the marked price of the article be ₹100.

$\therefore$ S.P. after the first discount = ₹$(100 - m)$ and discount at $n\%$ on ₹$(100 - m)$ = ₹$\dfrac{(100-m)\times n}{100}$.

$\therefore$ Single equivalent discount

$$= \left[m + \frac{(100-m)\times n}{100}\right]\%$$

$$= \left(\frac{100m + 100n - mn}{100}\right)\%$$

$$= \left(m + n - \frac{mn}{100}\right)\%$$

12 If three successive discounts on an article are *l%*, *m%* and *n%* respectively, then a single discount equivalent to the three successive discounts will be

$$\left[l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{100^2}\right]\%$$

Explanation

Let, the marked price of the article be ₹100.

∴ S.P. after the first discount = ₹(100 − *l*).

Second discount at *m*% on ₹(100 − *l*)

$$= ₹\frac{(100 - l) \times m}{100}$$

∴ S.P. after second the discount

$$= ₹(100 - l) - \frac{(100 - l)m}{100}$$

$$= ₹\frac{100\,(100 - l) - (100 - l)\,m}{100}$$

$$= ₹\frac{(100 - l) \cdot (100 - m)}{100}$$

Third discount at *n%* on $₹\frac{(100 - l)\,(100 - m)}{100}$

$$= ₹\frac{(100 - l)\,(100 - m)n}{100 \times 100}$$

∴ S.P. after the third discount

$$= ₹\frac{(100 - l)\,(100 - m)}{100} - \frac{(100 - l)\,(100 - m)n}{100 \times 100}$$

$$= ₹\frac{(100 - l)\,(100 - m)\,(100 - n)}{100 \times 100}$$

$$= \left(l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{(100)^2}\right)$$

∴ Single equivalent discount

$$= \left(l + m + n - \frac{(lm + ln + mn)}{100} + \frac{lmn}{(100)^2}\right)\%$$

Illustration 22: Find a single discount equivalent to two successive discounts of 10% and 20%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 - \frac{10 \times 20}{100}\right)\%, \text{ i.e., } 28\%$$

Illustration 23: Find out a single discount equivalent to three successive discounts of 10%, 20% and 30%.

Solution: The equivalent single discount is given by

$$\left(10 + 20 + 30 - \frac{(10 \times 20 + 10 \times 30 + 20 \times 30)}{100} + \frac{10 \times 20 \times 30}{100^2}\right)\%$$

i.e., $\left(60 - 11 + \frac{6}{10}\right)\% = \frac{496}{10}\%$ or 49.6%

Illustration 24: Two shopkeepers sell machines at the same list price. The first allows two successive discounts of 30% and 16% and the second 20% and 26%. Which discount series is more advantageous to the buyers?

Solution: A single discount equivalent to the two successive discounts of 30% and 16% is

$$\left(30 + 16 - \frac{30 \times 16}{100}\right)\%$$

or, $\left(46 - \frac{24}{5}\right)\%$ or, $41\frac{1}{5}\%$

Also, a single discount equivalent to the two successive discounts of 20% and 26% is

$$\left(20 + 26 - \frac{20 \times 26}{100}\right)\%$$

or, $\left(46 - \frac{26}{5}\right)\%$ or $40\frac{4}{5}\%$

Clearly, the discount series being offered by the first shopkeeper is more advantageous to the buyers.

13 A shopkeeper sells an item at ₹*z* after offering a discount of *d%* on labelled price. Had he not offered the discount, he would have earned a profit of *p%* on the cost price.

The cost price of each item is given by

$$\text{C.P.} = \left[\frac{100^2 z}{(100 - d)\,(100 + p)}\right].$$

Illustration 25: A shopkeeper sold sarees at ₹266 each after giving 5% discount on labelled price. Had he not given the discount, he would have earned a profit of 12% on the cost price. What was the cost price of each saree?

Solution: We have, labelled price *z* = ₹266, discount *d* = 5% and profit *p* = 12%

Using the formula

$$\text{C.P.} = \left[\frac{100^2 z}{(100 - d)\,(100 + p)}\right]$$

we get the cost price of each saree

$$= \left[\frac{100 \times 100 \times 266}{(100-5)(100+12)}\right]$$

$$= \frac{100 \times 100 \times 266}{95 \times 112} = ₹250.$$

EXERCISE-I

1. Mohan buys a watch for ₹350 and sells it for ₹392. Find out his percentage of profit.

(a) 9% (b) 12%
(c) 14% (d) None of these

2. Ramesh purchased a bicycle for ₹5200 and spent ₹800 on its repairs. He had to sell it for ₹5500. Find out his profit or loss per cent.

(a) $8\frac{1}{3}\%$ loss (b) $7\frac{1}{2}\%$ gain
(c) 9% (d) None of these

3. A man buys 10 articles for ₹8 and sells them at ₹1.25 per article. His gain per cent is:

(a) 55% (b) $56\frac{1}{4}\%$
(c) 40% (d) None of these

4. A toothpaste labeled at ₹80 is sold for ₹68. The rate of discount is:

(a) 12% (b) 14%
(c) 15% (d) None of these

5. Sardar Singh bought 200 dozen oranges at ₹10 a dozen. He spent ₹500 on transportation. He sold them at ₹1 each. What was his profit or loss per cent?

(a) 4% (b) 6%
(c) 5% (d) None of these

6. Mr Verma sold his scooter for ₹10500 at a gain of 5%. Find out the cost price of the scooter.

(a) ₹10300 (b) ₹10700
(c) ₹10000 (d) None of these

7. Suresh buys a camera for ₹1800 and sells it at 10% loss. Find out its selling price.

(a) ₹1620 (b) ₹1730
(c) ₹1650 (d) None of these

8. Hemant purchased 120 rims of paper at ₹80 per rim. He spent ₹280 on transportation, paid octroi at the rate of 40 paise per rim and paid ₹72 to the coolie. If he wants to have a gain of 8%, the selling price per rim must be:

(a) ₹89 (b) ₹90
(c) ₹95 (d) None of these

9. A shopkeeper loses 7% by selling a cricket ball for ₹31. For how much should he sell the ball so as to gain 5%?

(a) ₹50 (b) ₹65
(c) ₹35 (d) None of these

10. A shopkeeper sold some articles at ₹35 per piece and gained 40%. What would be the selling price of each article to earn 60% profit?

(a) ₹40 (b) ₹45
(c) ₹50 (d) None of these

11. A man bought apples at the rate of 6 for ₹20 and sold them at 4 for ₹16. His estimated profit % is:

(a) 23% (b) 18%
(c) 20% (d) None of these

12. A fruit vendor buys 10 bananas for ₹14 and sells them at 12 for ₹15. Find his percentage gain or loss.

(a) $10\frac{5}{7}\%$ loss (b) $10\frac{5}{9}\%$ gain
(c) 9% gain (d) None of these

13. If eggs are bought 12 for ₹10 and sold at 10 for ₹12. What is the gain or loss%.

(a) 40% loss (b) 44% gain
(c) 44% loss (d) None of these

14. If the cost price of 21 watches is equal to the selling price of 18 pieces, then what would be the gain per cent in this transaction?

(a) $6\frac{1}{2}\%$ (b) 7%
(c) $6\frac{2}{3}\%$ (d) None of these

15. A shopkeeper gains the cost of 8 metres thread by selling 40 metres thread. Find his gain per cent.

(a) 19% (b) 20%
(c) 22% (d) None of these

16. If the selling price of 2/3 of a certain quantity of milk be equal to the cost price of whole milk, then what will be the gain per cent in this transaction?

(a) 50% (b) 48%
(c) 53% (d) None of these

17. A shopkeeper sells 20 pencils for the same amount of money as he paid for 25 pencils. What is his gain per cent?

(a) 20% (b) 25%
(c) 24% (d) None of these

18. Mohit lost 18% by selling a bicycle for ₹1230. What per cent shall he gain or loss by selling it for ₹1600?

(a) $6\frac{2}{3}$% loss (b) 4% gain
(c) $6\frac{2}{3}$% gain (d) None of these

19. A shopkeeper sells an article at a gain of 10%. Had he sold it at a loss of 20%, its selling price would have been ₹180 less. What is the cost price of the article?

(a) ₹630 (b) ₹600
(c) ₹580 (d) None of these

20. A person sells 36 oranges per ₹1 and makes a loss of 4%. Find how many oranges per ₹ to be sold to have a gain of 8%?

(a) 1/32 (b) 5
(c) 1/16 (d) None of these

21. A person sells a colour TV at 10% below the cost price. Had he received ₹1494 more, he would have made a profit of $12\frac{1}{2}$%. What was the cost price of the colour TV?

(a) ₹6400 (b) ₹7200
(c) ₹6640 (d) None of these

22. Vijay sold a watch at a gain of 5%. Had he sold it for ₹72 more, he would have gained 13%. Find out the cost price of the watch.

(a) ₹900 (b) ₹910
(c) ₹870 (d) None of these

23. Sita sells a calculator to Gita at a gain of 17% and Gita sells it to Anu at a loss of 25%. If Anu pays ₹1842.75 for it, then what did Sita pay for it?

(a) ₹2080 (b) ₹2100
(c) ₹2110 (d) None of these

24. 'A' buys an article and sells it to 'B' at a profit of 10%, 'B' sells it to 'C' gaining 20%. If 'C' gives ₹924, what amount did 'A' give?

(a) ₹700 (b) ₹724
(c) ₹780 (d) None of these

25. 'A' sells an article to 'B' at a gain of 20% and 'B' sells it to 'C' at a gain of 10% and 'C' sells it to 'D' at a gain of $12\frac{1}{2}$%. If 'D' pays ₹29.70 what did it cost to 'A'?

(a) ₹20 (b) ₹24
(c) ₹18 (d) None of these

26. Rajesh sells taperecorder to Mihir at a loss of 10% and Mihir sells it to Shiv at a loss of 20%. If Shiv pays ₹1440 for it, at what price did Rajesh buy?

(a) ₹1920 (b) ₹2000
(c) ₹1800 (d) None of these

27. A man sells a scooter to his friend at 10% loss. If the friend sells it for ₹54000 and gains 20%, find out the original cost price of the scooter.

(a) ₹50000 (b) ₹45000
(c) ₹40000 (d) None of these

28. 'A' sells a good to 'B' at a profit of 10% and B sells it to 'C' at a profit of 20%. Find out the resultant profit.

(a) ₹35% (b) 20%
(c) ₹32% (d) None of these

29. A manufacturer sells an article to a wholesale dealer at a profit of 20%. The wholesale dealer sells it to a retail merchant at a loss of 5%. Find out the resultant profit or loss.

(a) 14% loss (b) 14% gain
(c) 12% gain (d) None of these

30. A man sold two watches for ₹3750 each. On one he gained 5%, and on the other, he lost 5%. What was his total gain or loss percentage?

(a) $1\frac{1}{4}$% (b) $\frac{1}{2}$%
(c) $\frac{1}{4}$% (d) None of these

31. A men sells two houses at the rate of ₹1.995 Lakhs each. On one house he gains 20% and on the other he loses 20%. His gain or loss per cent in the whole transaction is:

(a) 5% loss (b) 4%
(c) 4% loss (d) None of these

32. A shopkeeper sold two bicycles for ₹1500 each. On one, he gains 25% and on the other he loses 20%. His gain or loss per cent in the whole transaction is:

(a) $2\frac{18}{41}$% loss (b) $2\frac{18}{41}$% gain
(c) 2% gain (d) None of these

33. A man sells two articles, each for ₹640. He earns 20% profit on the first, and 40% profit on the second. Find his overall per cent profit.

(a) $29\frac{1}{2}$% (b) $28\frac{1}{2}$%
(c) $29\frac{3}{13}$% (d) None of these

34. A person sells two articles, each for ₹1040. He incurs 20% loss on the first and 10% loss on the second. Find out overall per cent loss.

(a) $12\frac{5}{17}$% (b) $15\frac{5}{17}$
(c) $13\frac{3}{4}$% (d) None of these

35. A grocer sells rice at a profit of 20% and uses a weight which is 25% less. Find out overall gain percentage.

(a) 60% (b) 65%
(c) 58% (d) None of these

36. A shopkeeper sells goods at 10% loss on cost price, but uses 20% less weight. What is his profit or loss percentage?

(a) 2% gain (b) $2\frac{1}{2}$% loss
(c) $2\frac{1}{2}$% gain (d) None of these

37. A cloth merchant says that due to slump in the market, he sells cloth at 10% loss, but he uses a inaccurate metre scale and actually gains 15%. Find out the actual length of the scale.

(a) 72.4 cm (b) 71.34 cm
(c) 78.25 cm (d) None of these

38. A cloth dealer professes to sell cotton at cost price, but uses a meter having a length of 80 cm only and charges for the meter. Find his gain per cent.

(a) 25% (b) 30%
(c) 40% (d) None of these

39. Sudeep buys two CDs for ₹380 and sells one at a loss of 22% and the other at a gain of 12%. If both the CDs are sold at the same price, then the cost price of two CDs is:

(a) ₹196, ₹225 (b) ₹230, ₹140
(c) ₹224, ₹156 (d) None of these

40. An article is listed at ₹65. A customer bought this article for ₹56.16 and received two successive discounts of which one is 10%. Find out the other discount in this discount scheme offered by the shopkeeper.

(a) 4% (b) 3%
(c) 6% (d) None of these

41. A cash payment that will settle a bill for 250 chairs at ₹50 per chair less 20% and 15% with a further discount of 5% on cash payment is:

(a) ₹7025 (b) ₹8075
(c) ₹8500 (d) None of these

42. A person sells taperecorders at ₹1134 each after giving a discount of 19% on the marked price. Had he not given the discount, he would have earned a profit of 40% on the cost price. The cost price of each taperecorder is:

(a) ₹1000 (b) ₹1200
(c) ₹1400 (d) None of these

EXERCISE-2
(BASED ON MEMORY)

1. Arun buys one kilogram of apples for ₹ 120 and sells it to Swati gaining 25%. Swati sells it to Divya who again sells it for ₹ 198, making a profit of 10%. What is the profit percentage made by Swati?

(a) 25% (b) 20%
(c) 16.67% (d) 15%

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2018]

2. A vendor buys bananas at 9 for ₹ 8 and sells at 8 for ₹ 9. What will be the profit or loss (in %)?

(a) 13.28% profit
(b) 26.56% loss
(c) 26.56% profit
(d) 13.28% loss

[SSC CGL Tier-II CBE, 2018]

3. If a stall sell a pizza at ₹ 200 he makes 20% loss if he wants to make 10% profit then at what price (in ₹) should he sell?

(a) 250 (b) 300
(c) 275 (d) 325

[SSC CGL Tier-II CBE, 2018]

4. If the selling price is tripled and cost price doubled the profit would become 65%. What is the present profit (in %)?

(a) 20 (b) 15
(c) 25 (d) 10

[SSC CGL Tier-II CBE, 2018]

5. If a vendor sells a coconut at ₹ 14.4 he makes 10% loss. If he wants to make 25% profit, there at what price (in ₹) should he sell?

(a) 18 (b) 20
(c) 16 (d) 22

[SSC CGL Tier-II CBE, 2018]

6. Selling price of a chair is ₹ 1386. If loss per cent is 23% then what is the cost price (in ₹) of chair?

(a) 1600 (b) 1800
(c) 1900 (d) 1067

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

7. Selling price of a glass is ₹ 1965 and loss per cent is 25%. If selling price is ₹ 3013, then what will be the profit per cent?

(a) 13 (b) 10.4
(c) 15 (d) 20

[SSC CHSL (10 + 2) Tier-I CBE, 2018]

8. On a certain item profit is 150%. If the cost price increase by 25% what will be the new profit margin (in %)

(a) 25 (b) 50
(c) 100 (d) 75

[SSC CGL Tier-II CBE, 2018]

9. A wholesaler had 200 dozens of mangoes. He sold some of these mangoes at 20% profit and the rest at 10% profit, so that he made 13% profit on selling all the mangoes. How many mangoes (in dozens) did he sells at 20% profit?

(a) 140 (b) 60
(c) 80 (d) 120

[SSC CGL Tier-II CBE, 2018]

10. At a village trade fair a man buys a horse and a camel together for ₹ 51,250. He sold the horse at a profit of 25% and the camel at a loss of 20%. If he sold both the animals at the same price, then the cost price of the cheaper animal was ₹ _____.

(a) 6600 (b) 7500
(c) 25000 (d) 20000

[SSC CGL Tier-II CBE, 2018]

11. Oil equal to 20% of the weight of ground nut is extracted in a mill. The matter left after extraction is sold as cattle feed at the rate of ₹ 12.5 per kg. The ground nuts are bought at ₹ 20 per kg. the processing cost is ₹ 5 per kg. At what price (₹ per kg) should the oil be sold to earn 20% profit on total costs (Total cost = cost of groundnuts and processing costs)?

(a) 250 (b) 150
(c) 200 (d) 100

[SSC CGL Tier-II CBE, 2018]

12. A man buys 10 oranges for a rupee and sells 8 oranges for a rupee. What is the profit per cent?

(a) 10 (b) 12.5
(c) 20 (d) 25

[SSC Multi-Tasking Staff, 2017]

13. A dishonest dealer defrauds to the extent of x% in buying as well as selling his goods by using faulty weight. What will be the gain per cent on his outlay?

(a) $2x$% (b) $\left(\frac{10}{x}+x^2\right)$%

(c) None of these (d) $\left(x+\frac{x^2}{100}\right)$%

[SSC CGL Tier-II (CBE), 2017]

14. A fan costing ₹ 1200 is being sold for ₹ 1500. What is the profit per cent?

(a) 10 (b) 12.5
(c) 25 (d) 20

[SSC Multi-Tasking Staff, 2017]

15. Raman sells a machine for ₹ 39 lakh at a loss. Had he sold it for ₹ 49 lakh, his gain would nhave been three times the loss. What is the cost price of the machine?

(a) ₹ 46.5 lakh (b) ₹ 62 lakh
(c) ₹ 36.5 lakh (d) ₹ 41.5 lakh

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

16. A rice trader buys 8 quintals of rice for ₹ 3,600, 10% rice is lost in transportation. At what rate should he sell to earn 15 % profit?

(a) ₹ 352.1 per quintal
(b) ₹ 517.5 per quintal
(c) ₹ 575 per quintal
(d) ₹ 582.3 per quintal

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

17. A rice trader buys 16 quintals of rice for ₹ 5632. 20% rice is lost in transportation. At what rate should he sell to earn 25 % profit?

(a) ₹ 225.2 per quintal
(b) ₹ 550 per quintal
(c) ₹ 440 per quintal
(d) ₹ 563.2 per quintal

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

18. Raheem sells a machine for ₹48 lakh at a loss. Had he sold it for ₹ 60 lakh, his gain would have been 5 times the former loss. Find the cost price of the machine.

(a) ₹ 58 lakh (b) ₹ 69.6 lakh
(c) ₹ 42 lakh (d) ₹ 50 lakh

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

19. A man purchased 120 reams of paper at ₹ 80 per ream. He spend ₹ 280 on transportation, paid octrol at the rate of 40 palse per ream and paid ₹ 72 to a porter. In order to gain 8% he must sell each ream of paper for:

(a) ₹ 90 (b) ₹ 87.5
(c) ₹ 89 (d) ₹ 85

[SSC Matric Level, 2017]

20. A man bought 15 mangoes for a rupee. How many mangoes were sold for a rupee so that there is a loss of 25%?

(a) 10 (b) 12
(c) 18 (d) 20

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. By selling a fan for ₹ 1900 a man has a loss of 5%, then at what price (in ₹) should he sell the fan to gain 20%?

(a) 2000 (b) 2400
(c) 2600 (d) 2800

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. Profit obtained on selling an article for ₹ 310 is equal to the loss incurred on selling that article for ₹ 230. What will be the loss percentage when selling price is ₹ 180?

(a) $16\frac{1}{3}$ (b) $16\frac{2}{3}$
(c) $33\frac{1}{3}$ (d) $33\frac{2}{3}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

23. A man bought nine pens for a rupee. How many pens should be sell for a rupee to gain 50%?

(a) 3 (b) 6
(c) 7 (d) 8

[SSC CAPFs ASI & Delhi Police SI, 2017]

24. A shopkeeper professes to sell his goods at cost price but uses a 930 gm weight instead of 1 kilogram weight. What will be the profit per cent of the shopkeeper?

(a) 3.76 (b) 7.52
(c) 9.85 (d) 10.36

[SSC CAPFs ASI & Delhi Police SI, 2017]

25. While selling an article for ₹ 18450, a person suffered a loss of 50%. At what price he should have sold the article (in ₹) to earn a profit of 50%?

(a) 13837 (b) 52000
(c) 55350 (d) 56775

[SSC CGL Tier-I (CBE), 2017]

26. A man gains 15% by selling a calculator for a certain price. If he sells it at the triple the price, then what will be the profit percentage?

(a) 125 (b) 175
(c) 225 (d) 245

[SSC CGL Tier-I CBE, 2017]

27. By selling 90 pens for ₹ 80 a man loses 20%. What should be the selling price (in ₹) of 90 pens for 20% profit?

(a) 90 (b) 100
(c) 110 (d) 120

[SSC CGL Tier-I CBE, 2017]

28. A trader buys 500kg of cotton for ₹ 9,000. 10% of this cotton is spoiled due to rain. At what rate (₹ / kg) should he sell the rest to earn 10% profit?

(a) 25 (b) 30
(c) 22 (d) 20

[SSC CGL Tier-I CBE, 2017]

29. A trader had 1200 kg of rice. He sold a part of it at 5% profit and the rest at 11% profit, so that he made a total profit of 7%. How much (in kg) rice did he sell at 5% profit?

(a) 900 (b) 600
(c) 400 (d) 800

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

30. If cost price is ₹ 150 less than the selling price and profit is 20%, then what is the selling price (in ₹)?

(a) 750 (b) 800
(c) 850 (d) 900

[SSC Multi-Tasking Staff, 2017]

31. If profit is $\left(\frac{1}{11}\right)$th of selling price, what is the profit per cent?

(a) $9\frac{1}{11}$ (b) 10

(c) $8\frac{1}{3}$ (d) $11\frac{1}{9}$

[SSC Multi-Tasking Staff, 2017]

32. The ratio of cost price and selling price of an article is 25 : 26. The per cent of profit will be

(a) 26% (b) 25%

(c) 1% (d) 4$

[SSC CGL Tier-II (CBE), 2017]

33. Profit of ₹ 144000 has to be divided among three partners Akram, Bipin and Chintan in the ratio 3 : 2 : 7. How much ₹ does Chinatn get?

(a) ₹ 84000 (b) ₹ 24000

(c) ₹ 36000 (d) ₹ 42000

[SSC CGL Tier-I (CBE), 2017]

34. *P* invests ₹ 9100 for 3 months. *Q* invests ₹ 6825 for 2 months and *R* ₹ 8190 for 5 months in a business. If the total profit amounts to ₹ 4158, how much profit should *Q* get?

(a) ₹ 682.50 (b) ₹ 693

(c) ₹ 1386 (d) ₹ 346.50

[SSC Matric Level, 2017]

35. Raman, Manan and Kamal are partners and invest in a business such that Raman invests $\frac{2}{5}$ th of total and Manan invests $\frac{3}{8}$th of the total. What is the ratio of profits of Raman, Manan and Kamal respectively?

(a) 16 : 15 : 9 (b) 16 : 15 : 31

(c) 2 : 3 : 5 (d) 15 : 16 :9

[SSC CAPFs ASI & Delhi Police SI, 2017]

36. If the ratio of the cost price and selling price of an article is 20 : 21, then what will be the profit per cent?

(a) 5% (b) 4.5%

(c) 6% (d) 7.5%

[SSC Multi-Tasking Staff, 2017]

37. On an article the profit is 210% of the cost price. If the cost price increase by 40% but the selling price remains constant, approximately what per cent of selling price will be the profit?

(a) 55 (b) 62

(c) 74 (d) 85

[SSC CAPFs ASI & Delhi Police SI, 2017]

38. The profit earned by a shopkeeper by selling a bucket at a gain of 8% is ₹ 28 more than when he sells it at a loss of 8%. The cost price (in Rupees) of the bucket is

(a) 170 (b) 190

(c) 175 (d) 165

[SSC CGL Tier-II CBE, 2017]

39. A trader sold an article at a gain of 20%. Had he purchased it for 40% more and sold for ₹ 24 less, he would have incurred a loss of 20%. What is the cost price (in ₹) of the article?

(a) 150 (b) 300

(c) 450 (d) 600

[SSC CAPFs ASI & Delhi Police SI, 2017]

40. Mr. Kapur purchased two toy cycles for ₹ 750 each. He sold these cycles, gaining 6% on one and losing 4% on the other. The gain or loss per cent in the whole transaction is

(a) 1% loss (b) 1% gain

(c) 1.5% loss (d) 1.5% gain

[SSC CGL Tier-II (CBE), 2017]

41. A man bought 500 metres of electronic wire at 50 paise per metre. He sold 50% of it at a profit of 5%. At what per cent should he sell the remainder so as to gain 10% on the whole transaction?

(a) 13% (b) 12.5%

(c) 15% (d) 20%

[SSC CGL Tier-II (CBE), 2017]

42. A shopkeeper sold one-third of his goods at a loss of 15%. To get a profit of 10% on the whole transaction, he should sell the remaining articles at a profit of

(a) $22\frac{1}{2}\%$ (b) $16\frac{2}{3}\%$

(c) 15% (d) 25%

[SSC Multi-Tasking Staff, 2017]

43. A trader buys two articles for ₹ 4000 each. While selling if he gains 12.5% on one and loses 20% on the other, then what will be the overall loss per cent?

(a) 2.5 (b) 3.75

(c) 5 (d) 5.25

[SSC CAPFs ASI & Delhi Police SI, 2017]

44. On selling an article for ₹170, a shopkeeper loses 15%. In order to gain 20%, he must sell that article at rupees:

(a) 215.50 (b) 212.50
(c) 240 (d) 210

[SSC CGL 2017]

45. By Selling 80 ball pens for ₹140, a retailer loses 30%. How many ball pens should he sell for ₹104 so as to make a profit of 30%?

(a) 32 (b) 52
(c) 48 (d) 42

[SSC CGL 2017]

46. If the selling price of 40 articles is equal to the cost price of 50 articles, the loss or gain per cent is

(a) 25% gain (b) 20% gain
(c) 25% loss (d) 20% loss

[SSC CGL Tier-I (CBE), 2016]

47. By what fraction selling price (S.P.) must be multiplied to get the cost price (C.P.) if the loss is 20%?

(a) $\frac{4}{5}$ (b) $\frac{8}{5}$
(c) $\frac{5}{4}$ (d) $\frac{6}{5}$

[SSC CGL Tier-II (CBE), 2016]

48. A shopkeeper sells rice at 10% profit and uses weight 30% less than the actual measure. His gain per cent is

(a) $57\frac{2}{3}\%$ (b) $57\frac{1}{7}\%$
(c) $57\frac{2}{5}\%$ (d) $57\frac{3}{7}\%$

[SSC CGL Tier-II (CBE), 2016]

49. A man bought 4 dozen eggs at ₹ 24 per dozen and 2 dozen eggs at ₹ 32 per dozen. To gain 20% on the whole, he should sell the eggs at

(a) ₹ 16 per dozen (b) ₹ 21 per dozen
(c) ₹ 32 per dozen (d) ₹ 35 per dozen

[SSC CGL Tier-I (CBE), 2016]

50. By selling cloth at ₹ 9 per meter, a shopkeeper loses 10%. Find the rate at which it should be sold so as to earn profit of 15%.

(a) ₹ 11.20 (b) ₹ 11.30
(c) ₹ 11.40 (d) ₹ 11.50

[SSC CGL Tier-I (CBE), 2016]

51. Kamal has some apples. He sold 40% more than he ate. If he sold 70 apples, how many dis he eat?

(a) 18 (b) 42
(c) 50 (d) 90

[SSC CGL Tier-I (CBE), 2016]

52. *A* and *B* jointly made a profit of ₹ 1650 and they decided to share it such that $\frac{1}{3}$ of *A*'s profit is equal to $\frac{2}{5}$ of *B*'s profit. Then profit of *B* is

(a) ₹ 700 (b) ₹ 750
(c) ₹ 850 (d) ₹ 800

[SSC CGL Tier-II CBE, 2016]

53. Anil started a business with an investment of ₹ 25,000. After 3 months, Vishal joined his business with a capital of ₹ 30, 000. At the end of the year, they have made a profit of ₹ 19,000 what will be Anil's share in the profit?

(a) ₹ 10,000 (b) ₹ 12,500
(c) ₹ 10,250 (d) ₹ 14,000

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

54. Instead of dividing 391 cookies among 3 children *A*, *B*, *C* in the ratio $\frac{1}{5}:\frac{1}{4}:\frac{1}{8}$, it was divided in to the ratio 5 : 4 : 8. Who gains the most and how many?

(a) *A*, 21 cookies
(b) *B*, 78 cookies
(c) *C*, 98 cookies
(d) *C*, 78 cookies

[SSC CPO SI & ASI, 2016]

55. After selling 5% of a quantity of sugar, 5 kg of sugar remains. Find the total quantity of sugar.

(a) 19 kg. (b) $5\frac{5}{19}$
(c) 100 kg (d) 95 kg

[SSC CGL Tier-I (CBE), 2016]

56. A shopkeeper purchases two items for ₹ 520. One of them is sold gaining 16% and the other at a loss of 10%, thus making no profit or loss. What is the selling price of the item sold at loss?

(a) ₹ 288 (b) ₹ 232
(c) ₹ 320 (d) ₹ 200

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2016]

57. The marked price of an article is ₹ 5000. But due to a special festive offer a certain per cent of discount is declared. Mr. X availed this opportunity and bought the article at reduced price. He then sold it at ₹ 5000

and thereby made a profit of $11\frac{1}{9}\%$. The percentage of discount allowed was

(a) 10 (b) $3\frac{1}{3}$

(c) $7\frac{1}{2}$ (d) $11\frac{1}{9}$

[SSC CGL Tier-II CBE, 2016]

58. A man buys a table and a chair for ₹ 500. He sells the table at a loss of 10% and the chair at a gain of 10%. He still gains ₹ 10 on the whole. The cost price of chair in rupees is:

(a) ₹ 200 (b) ₹ 250
(c) ₹ 300 (d) ₹ 350

[SSC CGL Tier-I (CBE), 2016]

59. If a commission at the rate of 10% is given to a bookseller on the marked price of a book by the publisher, the publisher gains 20%. If the commission is increased to 15% then the gain percent would be:

(a) $16\frac{2}{3}\%$ (b) $13\frac{1}{3}\%$

(c) $15\frac{5}{6}\%$ (d) $12\frac{1}{2}\%$

[SSC CPO SI, ASI, 2016]

60. 'A' bought a certain quantity of oranges at total cost of ₹1200/-. He sold 1/3rd of those oranges at 20% loss. If A earns an overall profit of 10%, at what percent profit did A sell the rest of the oranges?

(a) 16% (b) 15%
(c) 22% (d) 25%
(e) 20%

[IBPS, 2015]

61. Present age of Bob is equal to Abby's age 8 years ago. Four years hence, the respective ratio between Bob's age and Abby's age will be 4 : 5 at that time. What is Bob's present age?

(a) 24 years (b) 32 years
(c) 40 years (d) 20 years
(e) 28 years

[IBPS, 2015]

Directions (62-65): *In these questions, two equations numbered I and II are given. You have to solve both the equations and mark the appropriate option.*

Give Answer:

(1) If $x > y$

(2) If $x \geq y$

(3) If $x < y$

(4) If relationship between x and y cannot be determined

(5) $x \leq y$

62. I. $2x^2 + 19x + 45 = 0$

II. $2y^2 + 11y + 12 = 0$

[IBPS, 2015]

63. I. $3x^2 - 13x + 12 = 0$

II. $2y^2 - 15y + 28 = 0$

[IBPS, 2015]

64. I. $x^2 = 16$

II. $2y^2 - 17y + 36 = 0$

[IBPS, 2015]

65. I. $6x^2 + 19x + 15 = 0$

II. $3y^2 + 11y + 10 = 0$

[IBPS, 2015]

66. I. $2x^2 - 11x + 15 = 0$

II. $2y^2 - 11y + 14 = 0$

[IBPS, 2015]

67. 10% discount and then 20% discount in succession is equivalent to total discount of

(a) 15% (b) 30%
(c) 24% (d) 28%

[SSC, 2015]

68. The marked price of a watch was ₹720. A man bought the same for ₹550.80 after getting two successive discounts, the first being 10%. The second discount rate is

(a) 12% (b) 14%
(c) 15% (d) 18%

[SSC, 2015]

69. Allowing 20% and 15% successive discounts, the selling price of an article becomes ₹3,060; then the marked price will be

(a) ₹4,400 (b) ₹5,000
(c) ₹4,500 (d) ₹4,000

[SSC, 2015]

70. If a shopkeeper wants to give 20% discount on a toy, he has to sell it for ₹300. If he sells it at ₹405, then his gain percent is

(a) 5% (b) 8%
(c) 4% (d) 6%

[SSC, 2015]

71. Ram sold two horses at the same price. In one he gets a profit of 10% and in the other he gets a loss of 10%. Then Ram gets

(a) no loss or profit (b) 1% profit
(c) 1% loss (d) 2% loss

[SSC, 2015]

72. A manufacturer fixes his selling price at 33% over the cost of production. If cost of production goes up by 12% and manufacturer raises his selling price by 10%, his percentage profit is

(a) $36\frac{5}{9}\%$ (b) $28\frac{3}{8}\%$

(c) $30\frac{5}{8}\%$ (d) 35%

[SSC, 2015]

73. The marked price of a tape recorder is ₹12,600. A festival discount of 5% is allowed on it. Further for cash payment, a second discount of 2% is given. The cash payment, in rupees, is to be made for buying it is

(a) 11,370.60 (b) 11,073.60
(c) 11,703.60 (d) 11,730.60

[SSC, 2015]

74. A man purchases some oranges at the rate of 3 for ₹40 and the same quantity at 5 for ₹60. If he sells all the oranges at the rate of 3 for ₹50, find his gain or loss percent (to the nearest integer).

(a) 31% loss (b) 31% profit
(c) 34% loss (d) 32% profit

[SSC, 2015]

75. A dealer fixed the price of an article 40% above the cost of production. While selling it he allows a discount of 20% and makes a profit of ₹48. The cost of production (in ₹) of the article is

(a) 360 (b) 420
(c) 400 (d) 320

[SSC, 2015]

76. There would be a 10% loss, if rice is sold at ₹54 per kg. To earn a profit of 20%, the price of rice per kg will be

(a) ₹70 (b) ₹72
(c) ₹65 (d) ₹63

[SSC, 2015]

77. A man sells an article at 5% above its cost price. If he had bought it at 5% less than what he had paid for it and sold it at ₹2 less, he would have gained 10%. The cost price of the article is

(a) ₹400 (b) ₹100
(c) ₹200 (d) ₹300

[SSC, 2015]

78. Articles are marked at a price which gives a profit of 25%. After allowing a certain discount the profit reduces to 12½%. The discount percent is

(a) 12½% (b) 11.1%
(c) 10% (d) 12%

[SSC, 2015]

79. A shopkeeper gains 17% after allowing a discount of 10% on the marked price of an article. Find his profit percent if the article is sold at marked price allowing no discount.

(a) 27% (b) 30%
(c) 37% (d) 23%

[SSC, 2015]

80. List price of a book is ₹100. A dealer sells three such books for ₹274.50 after allowing discount at a certain rate. Find the rate of discount.

(a) 8.16% (b) 8.5%
(c) 8.34% (d) 8.33%

[SSC, 2015]

81. The printed price of an article is 40% higher than its cost price. Then the rate discount so that he gains 12% profit is:

(a) 15% (b) 21%
(c) 18% (d) 20%

[SSC, 2015]

82. Cost price of 100 books is equal to the selling price of 60 books. The gain percentage or less percentage is:

(a) $66\frac{2}{3}\%$ (b) $66\frac{3}{2}\%$
(c) 67% (d) 66%

[SSC, 2015]

83. If the discount of 10% is given on the marked price of a radio, the gain is 20%. If the discount is increased to 20%, the gain is:-

(a) $6\frac{1}{4}\%$ (b) 5%

(c) $7\frac{5}{8}\%$ (d) $6\frac{2}{3}\%$

[SSC, 2015]

84. A dealer buys an article listed at ₹100 and gets successive discounts of 10% and 20%. He spends 10% of the Cost Price on transportation. At what price should he sell the article to earn a profit of 15%?

(a) ₹92.00 (b) ₹91.08
(c) ₹90.80 (d) ₹91.20

[SSC, 2015]

85. A shopkeeper allows a discount of 10% on the marked price of a camera. Marked price of the camera, which costs him ₹600, to make a profit of 20% should be:-

(a) ₹700 (b) ₹800
(c) ₹750 (d) ₹650

[SSC, 2015]

86. A dealer sold a bicycle at a profit of 10%. Had he brought the bicycle at 10% less price and sold it at a price ₹60 more, he would have gained 25%. The cost price of the bicycle was-

(a) ₹2600 (b) ₹2000
(c) ₹2200 (d) ₹2400

[SSC, 2015]

87. The difference between successive discounts of 40% followed by 30% and 45% followed by 20% on the marked price of an article is ₹12. The marked price of the article is:

(a) ₹400 (b) ₹200
(c) ₹800 (d) ₹600

[SSC, 2015]

88. Find a simple discount equivalent to a discount series of 10%, 20% and 25%

(a) 45% (b) 55%
(c) 52% (d) 46%

[SSC, 2015]

89. Cost price of 100 books is equal to the selling price of 60 books. The gain or loss percentage will be:

(a) $66\frac{2}{3}\%$ (b) $66\frac{1}{4}\%$

(c) 66% (d) $66\frac{3}{4}\%$

[SSC, 2015]

90. An article which is marked at ₹975 is sold for ₹897. The % discount is?

(a) 6% (b) 10%
(c) 12% (d) 8%

[SSC, 2015]

91. A shopkeeper sold an item at 10% loss after giving a discount equal to half the marked price. The cost price of the item is:

(a) $\frac{1}{9}$ of marked price

(b) $\frac{4}{9}$ of marked price

(c) $\frac{5}{9}$ of marked price

(d) $\frac{7}{9}$ of marked price

[SSC, 2014]

92. A person purchased a saree for ₹7710 after availing a net discount of ₹1285. The percentage of discount, the saree-shop offers, is:

(a) $14\frac{1}{7}\%$ (b) $14\frac{2}{7}\%$

(c) $14\frac{3}{7}\%$ (d) $14\frac{4}{7}\%$

[SSC, 2014]

93. A cycle dealer offers a discount of 10% and still makes a profit of 26%. What does he pay for a cycle whose marked price is ₹840?

(a) ₹600 (b) ₹650
(c) ₹700 (d) ₹750

[SSC, 2014]

94. If the cost price of an item is $\frac{2}{5}$ of its marked price and if it is sold at a discount of 10%, then there will be:

(a) 25% profit (b) 40% profit
(c) 50% profit (d) 125% profit

[SSC, 2014]

95. An item costing ₹200 is being sold at 10% loss. If the price is further reduced by 5%, the selling price will be:

(a) ₹170 (b) ₹171
(c) ₹175 (d) ₹179

[SSC, 2014]

96. A shopkeeper buys 144 items at 90 paisa each. On the way 20 items are broken. He sells the remainder

at ₹1.20 each. His gain per cent correct to one place of decimal is:

(a) 13.8% (b) 14.6%
(c) 14.8% (d) 15.8%

[SSC, 2014]

97. There is a profit of 20% on the cost price of an article. The per cent of profit, when calculated on selling price is:

(a) $16\frac{2}{3}\%$ (b) 20%

(c) $33\frac{1}{3}\%$ (d) None of these

[SSC, 2014]

98. By selling an article for ₹102, there is a loss of 15%. When the article is sold for ₹134.40, the net result in the transaction is:

(a) 12% gain (b) 12% loss
(c) 10% loss (d) 15% gain

[SSC, 2014]

99. Two toys are sold at ₹504 each. One toy brings the dealer a gain of 12% and the other a loss of 4%. The gain or loss per cent by selling both the toys is:

(a) $3\frac{5}{13}\%$ Profit (b) $4\frac{5}{13}\%$ Profit

(c) $5\frac{1}{13}\%$ Profit (d) $2\frac{3}{13}\%$ Loss

[SSC, 2014]

100. A sold a horse to B for ₹4800 by losing 20%. B sells it to C at a price which would have given A a profit of 15%. B's gain is:

(a) ₹1800 (b) ₹1900
(c) ₹2000 (d) ₹2100

[SSC, 2014]

101. A reduction of 21% in the price of an item enables a person to buy 3 Kg more for ₹100. The reduced price of the item per Kg is:

(a) ₹5.50 (b) ₹7.50
(c) ₹10.50 (d) ₹7.00

[SSC, 2014]

102. A bakery bakes cake with the expectation that it will earna profit of 40% by selling each cake at marked price. But during the delivery to showroom 16% of the cakes were completely damaged and hence could not be sold. 24% of the cakes were slightly damaged and hence could be sold at 80% of the cost price. The remaining 60% of the cakes were sold at marked price. What is the percentage profit in the whole consignment?

(a) 3.2 (b) 2.4
(c) 2.8 (d) 4.2
(e) 3.6

[IBPS PO/MT, 2014]

103. The list price of a shirt is ₹440 and a customer pays ₹396 for it. The discount rate is

(a) 10 % (b) $10^{1/2}$ %
(c) 20 % (d) 12 %

[SSC, 2014]

104. A plate was sold for ₹6,300 after giving two successive discounts of $12^{1/2}$ % and 10 %. Find the marked price.

(a) ₹7,300 (b) ₹7,700
(c) ₹8,000 (d) ₹7,250

[SSC, 2014]

105. Nisha bought a number of oranges at 2 for a rupee and an equal number at 3 for a rupee. To make a profit of 20% she should sell a dozen for

(a) ₹6 (b) ₹8
(c) ₹10 (d) ₹12

[SSC, 2014]

106. A shopkeeper listed the price of goods at 30% above the cost price. He sells half the stock at this price, one fourth of the stock at a discount of 15% and the remaining at 30% discount. His overall profit is

(a) $15\frac{3}{8}\%$ (b) 15 %

(c) $-15\frac{3}{5}\%$ (d) $15\frac{2}{3}\%$

[SSC, 2014]

107. A trader buys good at 20% discount on marked price. If he wants to make a profit of 25% after allowing a discount of 20%, by what percent should his marked price be greater than the original marked price?

(a) 15 % (b) 65 %
(c) 25 % (d) 20 %

[SSC, 2014]

108. A shopkeeper allows a discount of 10% on the marked price of an item but charges a sales tax of 8% on the discounted price. If the customer pays ₹3,402 as the price including the sales tax, then the marked price is

(a) ₹3,400 (b) ₹3,500
(c) ₹3,600 (d) ₹3,800

[SSC, 2014]

109. A fruit-seller buys some oranges and by selling 40% of them he fealises the cost price of all the oranges. As the oranges being to grow over-ripe, he reduces the price and sells 80% of the remaining oranges at half the previous rate of profit. The rest of the oranges being rotten are thrown away. The overall percentage of profit is

(a) 80 (b) 84
(c) 94 (d) 96

[SSC, 2014]

110. A merchant advertises 10% off on the items bought from his store. The total discount got by a customer who bought a cooker worth \$650, a heater worth \$500 and a bag worth \$65 is

(a) \$128.50 (b) \$121.50
(c) \$120.50 (d) \$123.50

[SSC, 2014]

111. By selling an article for \$450, a man loses 10%. The gain or loss percent if he sells it for \$540 is

(a) loss 9% (b) gain 8%
(c) loss 8% (d) gain 9%

[SSC, 2014]

112. A man loses $20\frac{1}{2}$ % of his money and after spending 80% of the remainder, he is left with \$159. How much did he have at first?

(a) \$1,000 (b) \$1,200
(c) \$500 (d) \$800

[SSC, 2014]

113. The cost price of a table is ₹3,200. A merchant wants to make 25% profit by selling it. At the time of sale he declares a discount of 20% on the marked price. The marked price (in ₹) is

(a) 4,000 (b) 4,500
(c) 5,000 (d) 6,000

[SSC, 2014]

114. A shopkeeper allows a discount of 12.5% on the marked price of a certain article and makes a profit of 20%. If the article costs the shopkeeper ₹210, then the marked price of the article will be

(a) ₹386 (b) ₹288
(c) ₹387 (d) ₹350

[SSC, 2014]

115. A businessman allows a discount of 10% on the marked price. What percent above the cost price must he mark his goods to make a profit of 17%?

(a) 30% (b) 20%
(c) 27% (d) 18%

[SSC, 2014]

116. Charging 30% above its production cost a radio maker puts a label of ₹286 on a ratio as its price. But at the time of selling it, he allows 10% discount on the labeled price. What will his gain be?

(a) ₹198 (b) ₹37.40
(c) ₹257.40 (d) ₹254.40

[SSC, 2014]

117. A tradesman marks his goods 30% more than the cost price. If he allows a discount of $6\frac{1}{4}$% then his gain per cent is:

(a) $23\frac{3}{4}$% (b) 22%

(c) $21\frac{7}{8}$% (d) 30%

[SSC, 2013]

118. A shopkeeper purchased a chair marked at ₹600 at two successive discounts of 15% and 20%. He spent ₹28 on transportation and sold the chair for ₹545. His gain per cent was:

(a) 25% (b) 30%
(c) 35% (d) 20%

[SSC, 2013]

119. The marked price of a piano was ₹15,000. At the time of sale, there were successive discounts of 20% 10% and 10% on it. The sale price was:

(a) ₹9,720 (b) ₹9,750
(c) ₹9,760 (d) ₹9,780

[SSC, 2013]

120. By selling 25 metres of cloth a trader gains the selling price of 5 metres of cloth. The gain of the trader in % is:

(a) 25 (b) 20
(c) 28 (d) 29

[SSC, 2013]

121. A sells a suitcase to B at 10% profit. B cells it to C at 30% profit. If C pays ₹2,860 for it, then the price at which A bought it is:

(a) ₹1,000 (b) ₹1,600
(c) ₹2,000 (d) ₹2,500

[SSC, 2013]

122. Gita buys a plot of land for ₹96,000. She sells $\frac{2}{5}$ of it at a loss of 6%. She wants to make a profit of 10% on the whole transaction by selling the remaining

land. The gain percentage on the remaining land is:

(a) 20 (b) $20\frac{2}{3}$

(c) 14 (d) 7

[SSC, 2013]

123. An article is sold at a gain of 15%. Had it been sold for ₹27 more, the profit would have been 20%. The cost price of the article is:

(a) ₹500 (b) ₹700

(c) ₹540 (d) ₹545

[SSC, 2013]

124. On selling 17 balls at₹720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is:

(a) 45 (b) 50

(c) 55 (d) 60

[SSC, 2013]

125. Two items A and B are sold at a profit of 10% and 15%, respectively. If the amount of profit received is the same, then the cost price of A and B may be:

(a) ₹1,000, ₹1,500 (b) ₹5,000, ₹2,000

(c) ₹3,000, ₹2,000 (d) ₹3,000, ₹5,000

[SSC, 2013]

126. Arun marks up the computer he is selling by 20% profit and sells them at a discount of 15%. Arun's net gain per cent is:

(a) 4 (b) 2

(c) 3.5 (d) 2.5

[SSC Assistant Grade III, 2013]

127. A dealer buys a table listed at ₹1,500 and gets successive discounts of 20% and 10%. He spends ₹20 on transportation and sells at a profit of 20%. Find the selling price of the table.

(a) ₹1,320 (b) ₹1,080

(c) ₹1,200 (d) ₹1,230

[SSC Assistant Grade III, 2013]

128. A sells an article to B at a gain of 20% and B sells it to C at a gain of 10% and C sells it to D at a gain of $12\frac{1}{2}\%$. If D pays ₹29.70, then A purchased the article for:

(a) ₹40 (b) ₹10

(c) ₹20 (d) ₹30

[SSC Assistant Grade III, 2013]

129. By selling 80 ball pens for ₹140 a retailer loses 30%. How many ball pens should he sell for ₹104 so as to make a profit of 30%?

(a) 32 (b) 52

(c) 48 (d) 42

[SSC Assistant Grade III, 2013]

130. A shopkeeper sells two watches for ₹308 each. On one watch he earns 12% profit and on the others he suffers 12% loss. His profit or loss in the entire transaction was:

(a) $1\frac{11}{25}\%$ loss (b) $1\frac{11}{25}\%$ gain

(c) $3\frac{2}{25}\%$ loss (d) $3\frac{2}{25}\%$ gain

(e) None of these

[IBPS PO/MT, 2013]

131. A discountof 40% on the marked price of atrouser enables Ajit to purchase a shirt also which costs him ₹320. How much did Ajit payfor the trouser?

(a) ₹ 480 (b) ₹ 540

(c) ₹ 800 (d) ₹ 400

[SSC Assistant Grade III, 2012]

132. Rahim bought a gift item for ₹510 after getting a discount of 15%. He then sells it 5% above the marked price. The profit earned in this deal is:

(a) ₹150 (b) ₹120

(c) ₹100 (d) ₹90

[SSC Assistant Grade III, 2012]

133. A shopkeeper marks his goods at 40% above their cost price. He is able to sell $\frac{3}{4}$th of his goods at this price, and the remaining at 40% discount. Assuming that the shopkeeper is able to sell the goods he buys, find his loss or gain as % of the whole transaction.

(a) 20% loss (b) 23% loss

(c) 26% gain (d) 30% gain

[SSC Assistant Grade III, 2012]

134. A fruit seller bought 240 bananas at the rate of ₹48 per dozen. He sells half of them at the rate of ₹5 per banana, $\frac{1}{6}$th of the remaining are found to be rotten. The price per banana at which he has to sell the remaining bananas to get a profit of 25% on his entire investment is:

(a) ₹5.5 (b) ₹6.0

(c) ₹5.0 (d) ₹6.5

[SSC Assistant Grade III, 2012]

135. A and B started a business by investing ₹3,50,000 and ₹1,40,000 respectively. A gets 20% of the yearly profit for managing the business. Thereafter the profit is divided in the ratio of the capital. If A receives

totally ₹38,000 more than B at the end of a year, then the profit is:
(a) ₹28,000 (b) ₹2,80,000
(c) ₹1,05,000 (d) ₹70,000

[SSC, 2012]

136. A fan in a shop is offered at a discount of 10%. It is sold during clearance sale at 6% discount over the already discounted price at ₹846. The original marked price of the fan is:
(a) ₹1000 (b) ₹900
(c) ₹850 (d) ₹896

[SSC, 2012]

137. A trader allows a trade discount of 20% and a cash discount of $6\frac{1}{4}\%$ on the marked price of the goods and gets a net gain of 20% of the cost. By how much above the cost should the goods be marked for the sale?
(a) 40% (b) 50%
(c) 60% (d) 70%

[SSC, 2012]

138. A discount series of 10%, 20% and 40% is equal to a single discount of:
(a) 56.80% (b) 50%
(c) 70% (d) 43.20%

[SSC, 2012]

139. Tarun bought a TV with 20% discount on the labelled price. Had he bought it with 25% discount, he would have saved ₹500. At what price did he buy the TV?
(a) ₹7,500 (b) ₹8,500
(c) ₹8,000 (d) ₹7,400

[SSC, 2012]

140. A manufacturer sells an article to a wholesale dealer at a profit of 10%. The wholesale dealer sells it to a shopkeeper at 20% profit. The shopkeeper sells it to a customer for ₹56,100 at a loss of 15%. Then the cost price of the article to the manufacturer is:
(a) ₹25,000 (b) ₹10,000
(c) ₹50,000 (d) ₹55,000

[SSC, 2012]

141. A loss of 19% gets converted into a profit of 17% when the selling price is increased by ₹162. The cost price of the article is:
(a) ₹450 (b) ₹600
(c) ₹360 (d) ₹540

[SSC, 2012]

142. A man purchased 150 pens at the rate of ₹12 per pen. He sold 50 pens at gain of 10%. The percentage gain at which he must sell the remaining pens so as to gain 15% on the whole outlay is:
(a) $21\frac{1}{2}\%$ (b) 20%
(c) 17% (d) $17\frac{1}{2}\%$

[SSC, 2012]

143. A dealer sold two types of goods for ₹10,000 each. On one of them, he lost 20% and on the other he gained 20%. His gain or loss per cent in the entire transaction was:
(a) 2% loss (b) 2% gain
(c) 4% gain (d) 4% loss

[SSC, 2012]

144. The cost price of 40 articles is the same as the selling price of 25 articles. Find the gain per cent.
(a) 65% (b) 60%
(c) 15% (d) 75%

[SSC, 2012]

145. A sells an article to B making a profit of $\frac{1}{5}th$ his outlay. B sells it to C, gaining 20%. If C sells it for ₹600 and incurs a loss of $\frac{1}{6}th$ his outlay, the cost price of A is:
(a) ₹600 (b) ₹500
(c) ₹720 (d) ₹800

[SSC, 2012]

146. A man had a certain amount with him. He spent 20% of that to buy an article and 5% of the remaining on transport. Then he gifted ₹120. If he is left with ₹1,400, the amount he spent on transport is:
(a) ₹76 (b) ₹61
(c) ₹95 (d) ₹80

[SSC, 2012]

147. An article was purchased for ₹78,350. Its price was marked up by 30%. It was sold at a discount of 20% on the marked-up price. What was the profit per cent on the cost price?
(a) 4% (b) 7%
(c) 5% (d) 3%
(e) 6%

[IBPS PO/MT, 2012]

148. 20% loss on selling price is what per cent loss on the cost price?
(a) 25% (b) $16\frac{2}{3}\%$
(c) 15% (d) $16\frac{1}{3}\%$

[SSC (GL), 2011]

149. X sells two articles for ₹4,000 each with no loss and no gain in the interaction. If one was sold at a gain of 25% the other is sold at a loss of:

(a) 25% (b) $18\frac{2}{9}\%$

(c) $16\frac{2}{3}\%$ (d) 20%

[SSC (GL), 2011]

150. A man purchased some eggs at 3 for ₹5 and sold them at 5 for ₹12. Thus, he gained ₹143 in all. The number of eggs he bought is:

(a) 210 (b) 200

(c) 195 (d) 190

[SSC (GL), 2011]

151. The cost price of an article is 64% of the marked price. The gain percentage after allowing a discount of 12% on the marked price is:

(a) 37.5% (b) 48%

(c) 50.5% (d) 52%

[SSC (GL), 2011]

152. By selling an article for ₹144, a person gained such that the percentage gain equals the cost price of the article. The cost price of the article is:

(a) ₹90 (b) ₹80

(c) ₹75 (d) ₹60

[SSC (GL), 2011]

153. A man sells two article for ₹5000 each neither losing nor gaining in the deal. If he sold one of them at a gain of 25%, the other article is sold at a loss of:

(a) $15\frac{2}{3}\%$ (b) $16\frac{2}{3}\%$

(c) $17\frac{1}{3}\%$ (d) $18\frac{1}{3}\%$

[SSC (GL), 2011]

154. A man bought orange at the rate of 8 for ₹34 and sold them at the rate of 12 for ₹57. How many oranges should be sold to earn a net profit of ₹45?

(a) 90 (b) 100

(c) 135 (d) 150

[SSC (GL), 2011]

155. A shopkeeper allows 23% commission on his advertised price and still makes a profit of 10%. If he gains ₹56 on one item. His advertised price of the item, in ₹, is:

(a) 820 (b) 780

(c) 790 (d) 800

[SSC (GL), 2011]

156. Seema purchased an item for ₹9600 and sold it for loss of 5 per cent. From that money she purchased another item and sold it for gain of 5 per cent. What is her overall gain/loss?

(a) Loss of ₹36 (b) Profit of ₹24

(c) Loss of ₹54 (d) None of these

[Bank of Baroda PO Examination, 2011]

157. By selling an article at $\frac{3}{4}th$ of the marked price, there is gain of 25%. The ratio of the marked price and the cost price is:

(a) 5:3 (b) 3:5

(c) 3:4 (d) 4:3

[SSC, 2011]

158. Successive discounts of 10%, 20% and 50% will be equivalent to a single discount of:

(a) 36% (b) 64%

(c) 80% (d) 56%

[SSC, 2011]

159. A retailer offers the following discount schemes for buyers on an article:

I. Two successive discounts of 10%.

II. A discount of 12% followed by a discount of 8%.

III. Successive discounts of 15% and 5%.

IV. A discount of 20%.

The selling price will be minimum under the scheme

(a) I (b) II

(c) III (d) IV

[SSC, 2011]

160. The value of an article depreciates every year at the rate of 10% of its value. If the present value of the article is ₹729, then its worth 3 years ago was:

(a) ₹1250 (b) ₹1000

(c) ₹1125 (d) ₹1200

[SSC, 2011]

161. Nitin bought some oranges at ₹40 a dozen and an equal number at ₹30 a dozen. He sold them at ₹45 a dozen and made a profit of ₹480. The number of oranges, he bought, was:

(a) 48 (b) 60

(c) 72 (d) 84

[SSC, 2011]

162. A man buys two chairs for a total cost of ₹900. By selling one for $\frac{4}{5}$ of its cost and the other for $\frac{5}{4}$

of its cost, he makes a profit of ₹90 on the whole transaction. The cost of the lower priced chair is:

(a) ₹360 (b) ₹400
(c) ₹420 (d) ₹300

[SSC, 2011]

163. By selling 100 oranges, a vendor gains the selling price of 20 oranges. He gain per cent is:

(a) 20 (b) 25
(c) 30 (d) 32

[SSC, 2011]

164. 60% of the cost price of an article is equal to 50% of its selling price. Then the percentage of profit or loss on the cost price is:

(a) 20% loss (b) $16\frac{2}{3}$% profit
(c) 20% profit (d) 10% loss

[SSC, 2011]

165. Maninder bought two horses at ₹40,000 each. He sold one horse at 15% gain, but had to sell the second horse at a loss. If he had suffered a loss of ₹3,600 on the whole transaction, then the selling price of the second horse is:

(a) ₹30,000 (b) ₹30,200
(c) ₹30,300 (d) ₹30,400

[SSC, 2011]

166. A fruit-seller buys x guavas for ₹y and sells y guavas for ₹x. If $x>y$, then he made:

(a) $\frac{x^2-y^2}{xy}$% loss (b) $\frac{x^2-y^2}{xy}$% gain
(c) $\frac{x^2-y^2}{y^2}$% loss (d) $\frac{x^2-y^2}{y^2}\times 100$% gain

[SSC, 2011]

167. Profit earned by an organization is distributed among officers and clerks in the ratio of 5:3. If the number of officers is 45 and the number of clerks is 80 and the amount received by each officer is ₹25,000, what was the total amount of profit earned?

(a) ₹22 Lakhs (b) ₹18.25 Lakhs
(c) ₹18 Lakhs (d) ₹23.25 Lakhs
(e) None of these

[SBI Associates Banks PO, 2011]

168. A shopkeeper labeled the price of his articles in order to earn a profit of 30% on the cost price. He, then, sold the articles by offering a discount of 10% on the labelled price. What is the actual rate of profit he earned in the deal?

(a) 18% (b) 8%
(c) 20% (d) Cannot be determined
(e) None of these

[SBI Associates Banks PO, 2011]

169. Kamya purchased an item for ₹46,000 and sold it at aloss of 12 per cent. With that amount she purchased another item which he sold at a gain of 12%. What was her overall gain/loss?

(a) Loss of ₹662.40 (b) Profit of ₹662.40
(c) Loss of ₹642.80 (d) Profit of ₹642.80
(e) None of these

[Allahabad Bank PO, 2011]

170. Rehaan purchased a bike for ₹54,000. He sold it at a loss of 8%. With that money, he again purchased another bike and sold that at a profit of 10 per cent. What is his overall loss/profit?

(a) loss ₹657 (b) profit ₹567
(c) loss ₹648 (d) profit₹648
(e) None of these

[Corporation Bank PO, 2011]

171. A shopkeeper earns a profit of 12% on selling a book at 10% discount on the printed price. The ratio of the cost price and the printed price of the book is:

(a) 45:56 (b) 45:51
(c) 47:56 (d) 47:51

[SSC (GL), 2010]

172. What profit/loss per cent did Ravi earn if he purchased an item of ₹5600 and sold it at 3/4 of its cost price?

(a) Loss of 20 per cent
(b) Gain of 25 per cent
(c) Neither gain nor loss
(d) None of these

[OBC PO Examination, 2010]

173. A shopkeeper sells notebooks at the rate of ₹45 each and earns a commission of 4%. He also sells pencil box at the rate of ₹80 each and earns a commission of 20%. How much amount of commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day?

(a) ₹1956 (b) ₹1586
(c) ₹1496 (d) ₹1596
(e) None of these

[CBI PO Examination, 2010)]

174. A shopkeeper bought 30 Kg of wheat at the rate of ₹45 per Kg. He sold 40% of the total quantity at the rate of ₹50 per Kg. Approximately, at what price per Kg should he sell the remaining quantity to make 25 per cent overall profit ?

(a) ₹54 (b) ₹52
(c) ₹50 (d) ₹60

[Allahabad Bank PO Examination, 2010]

175. A person sold a horse at a gain of 15%. Had he bought it for 25% less and sold it for ₹60 less, he would have made a profit of 32%. The cost price of the horse was:

(a) ₹370 (b) ₹372
(c) ₹375 (d) ₹378

[SSC, 2010]

176. A sells an article to B at a gain of 25%, B sells it to C at a gain 20% and C sells it to D at a gain of 10%. If D pays ₹330 for it, how much did it cost A?

(a) ₹200 (b) ₹250
(c) ₹275 (d) ₹290

[SSC, 2010]

177. By selling an article for ₹21, a man lost such that the percentage loss was equal to the cost price. The cost price of the article was:

(a) ₹30 or ₹70 (b) ₹35 or ₹60
(c) ₹45 (d) ₹50

[SSC, 2010]

178. Half of 100 articles were sold at a profit of 20% and the rest at a profit of 40%. If all of the articles had been sold at a profit of 25%, the total profit would have been ₹100 less than earlier profit. The cost price of each article was:

(a) ₹10 (b) ₹15
(c) ₹20 (d) ₹30

[SSC, 2010]

179. The marked price of a clock is ₹3200. It is to be sold at ₹2448 at two successive discounts. If the first discount is 10%, then the second discount is:

(a) 5% (b) 10%
(c) 15% (d) 20%

[SSC, 2010]

180. A dealer marks his goods 30% above his cost price and then allows 15% discount on it. What is the cost price of an article on which he gains ₹84?

(a) ₹800 (b) ₹560
(c) ₹373.33 (d) ₹280

[SSC, 2010]

181. A shopkeeper wishes to give 5% commission on the marked price of an article but also wants to earn a profit of 10%. If his cost price is ₹95, then the marked price is:

(a) ₹100 (b) ₹110
(c) ₹120 (d) ₹130

[SSC, 2010]

182. A shopkeeper sells sugar in such a way that the selling price of 950 g of sugar is the same as the cost price of 1 Kg of sugar. What is his gain per cent?

(a) $5\frac{5}{19}$ (b) $5\frac{1}{5}$
(c) 5 (d) $4\frac{1}{19}$

[SSC, 2010]

183. A person bought a horse and a carriage for ₹20000. Later, he sold the horse at 20% profit and the carriage at 10% loss. Thus, he gained 2% in the whole transaction. The cost price of the horse was:

(a) ₹7200 (b) ₹7500
(c) ₹8000 (d) ₹9000

[SSC, 2010]

184. A sells an article to B at 15% profit. B sells it to C at 10% loss. If C pays ₹517.50 for it then A purchased it at:

(a) ₹500 (b) ₹750
(c) ₹1000 (d) ₹1250

[SSC, 2010]

185. An article is sold at a certain fixed price. By selling it at $\frac{2}{3}$ of that price, one loses 10%. The gain per cent on selling it at the original price is:

(a) 20 (b) $33\frac{1}{3}$
(c) 35 (d) 40

[SSC, 2010]

186. A sells an article to B for ₹45,000 losing 10% in the transaction. B sells it to C at a price which would have given a profit of 10% to A. By what per cent does B gain?

(a) $\frac{75}{2}$ (b) $\frac{100}{3}$
(c) $\frac{200}{9}$ (d) $\frac{150}{7}$

[SSC, 2010]

187. The cost price of an article is 80% of its marked price for sale. How much per cent does the tradesman gain after allowing a discount of 12%?

(a) 20 (b) 12
(c) 10 (d) 8

[SSC, 2010]

188. A merchant purchases a wrist watch for ₹450 and fixes its list price in such a way that after allowing a discount of 10%, he earns a profit of 20%. Then the list price (in rupees) of the wrist watch is:

(a) ₹500 (b) ₹600
(c) ₹750 (d) ₹800

[SSC, 2010]

189. A, B, C are partners in a business. During a particular year, A received one third of the profit, B received $\frac{1}{4}$ of the profit and C received the remaining ₹5000. How much amount of money did A receive?

(a) ₹1000 (b) ₹3000
(c) ₹4000 (d) ₹5000

[SSC, 2010]

190. A shopkeeper sells notebooks at the rate of ₹457 each and earns a commission of 4%. He also sells pencil boxes at the rate of ₹80 each and earns a commission of 20%. How much amount of commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day?

(a) ₹1,956 (b) ₹1,586
(c) ₹1,496 (d) ₹1,596
(e) None of these

[CBI PO, 2010]

191. Meenal purchased a car for ₹2,50,000 and sold it for ₹3,48,000. What is the per cent profit she made on the car?

(a) 40 (b) 39.2
(c) 38.4 (d) 38
(e) None of these

[Corporation Bank PO, 2010]

192. A shopkeeper bought 30 Kg of wheat at the rate of ₹45 per Kg. He sold 40% of the total quantity at the rate of ₹50 per Kg. Approximately, at what price per Kg should he sell the remaining quantity to make 25 per cent overall profit?

(a) ₹54 (b) ₹52
(c) ₹50 (d) ₹60
(e) ₹56

[Allahabad Bank PO, 2010]

ANSWER KEYS

EXERCISE-1

1. (b)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (c)	**7.** (a)	**8.** (b)	**9.** (c)	**10.** (a)	**11.** (c)	**12.** (a)	**13.** (b)
14. (c)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (a)	**21.** (c)	**22.** (a)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (b)
27. (a)	**28.** (c)	**29.** (b)	**30.** (c)	**31.** (c)	**32.** (a)	**33.** (c)	**34.** (b)	**35.** (a)	**36.** (c)	**37.** (c)	**38.** (a)	**39.** (c)
40. (a)	**41.** (b)	**42.** (a)										

EXERCISE-2

1. (b)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (d)	**11.** (d)	**12.** (d)	**13.** (c)
14. (c)	**15.** (d)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (d)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (b)	**25.** (c)	**26.** (d)
27. (d)	**28.** (c)	**29.** (d)	**30.** (d)	**31.** (b)	**32.** (d)	**33.** (a)	**34.** (b)	**35.** (a)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (b)
40. (b)	**41.** (c)	**42.** (a)	**43.** (b)	**44.** (c)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (b)	**49.** (c)	**50.** (d)	**51.** (c)	**52.** (b)
53. (a)	**54.** (c)	**55.** (b)	**56.** (a)	**57.** (a)	**58.** (c)	**59.** (b)	**60.** (d)	**61.** (b)	**62.** (c)	**63. (4)**	**64.** (e)	**65.** (b)
66. (d)	**67.** (d)	**68.** (c)	**69.** (c)	**70.** (b)	**71.** (a)	**72.** (c)	**73.** (d)	**74.** (b)	**75.** (c)	**76.** (b)	**77.** (a)	**78.** (b)
79. (b)	**80.** (b)	**81.** (d)	**82.** (a)	**83.** (d)	**84.** (b)	**85.** (b)	**86.** (d)	**87.** (d)	**88.** (d)	**89.** (a)	**90.** (d)	**91.** (c)
92. (b)	**93.** (a)	**94.** (d)	**95.** (b)	**96.** (c)	**97.** (a)	**98.** (a)	**99.** (a)	**100.** (d)	**101.** (d)	**102.** (a)	**103.** (a)	**104.** (c)
105. (a)	**106.** (a)	**107.** (c)	**108.** (b)	**109.** (b)	**110.** (b)	**111.** (b)	**112.** (a)	**113.** (c)	**114.** (b)	**115.** (a)	**116.** (b)	**117.** (c)
118. (a)	**119.** (a)	**120.** (a)	**121.** (c)	**122.** (b)	**123.** (c)	**124.** (d)	**125.** (c)	**126.** (b)	**127.** (a)	**128.** (c)	**129.** (a)	**130.** (a)
131. (a)	**132.** (b)	**133.** (c)	**134.** (b)	**135.** (d)	**136.** (a)	**137.** (c)	**138.** (a)	**139.** (c)	**140.** (c)	**141.** (a)	**142.** (d)	**143.** (d)
144. (b)	**145.** (b)	**146.** (d)	**147.** (a)	**148.** (b)	**149.** (d)	**150.** (c)	**151.** (a)	**152.** (b)	**153.** (b)	**154.** (a)	**155.** (d)	**156.** (d)
157. (a)	**158.** (b)	**159.** (d)	**160.** (b)	**161.** (a)	**162.** (d)	**163.** (b)	**164.** (c)	**165.** (d)	**166.** (d)	**167.** (d)	**168.** (e)	**169.** (a)
170. (d)	**171.** (a)	**172.** (d)	**173.** (d)	**174.** (d)	**175.** (c)	**176.** (a)	**177.** (a)	**178.** (c)	**179.** (c)	**180.** (a)	**181.** (b)	**182.** (a)
183. (c)	**184.** (a)	**185.** (c)	**186.** (c)	**187.** (c)	**188.** (b)	**189.** (c)	**190.** (e)	**191.** (b)	**192.** (d)			

EXPLANATORY ANSWERS

EXERCISE-1

1. (b) Here, C.P. = ₹350, S.P. = ₹392

Profit = S.P. − C.P. = 392 − 350 = ₹42.

$\therefore$ Profit % $= \frac{\text{Profit} \times 100}{\text{C.P.}} = \frac{42 \times 100}{350} = 12\%$

2. (a) C.P. of the bicycle = 5200 + 800 = ₹6000

S.P. = ₹5500. Since S.P. < C.P.,

$\therefore$ Loss = C.P. − S.P. = 6000 − 5500 = ₹500.

$\therefore$ Loss % $= \frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{500 \times 100}{6000}$

$= \frac{25}{3}\%$ or $8\frac{1}{3}\%$

3. (b) Cost price of 10 articles (C.P.) = ₹8.

Selling price of 10 articles (S.P.) = 1.25 × 10

= ₹12.50

Profit = S.P. − C.P. = 12.50 − 8 = ₹4.50.

$\therefore$ Gain % = $\frac{\text{Gain} \times 100}{\text{C.P}} = \frac{4.50 \times 100}{8} = 56\frac{1}{4}\%$

4. (c) Marked price (M.P.) = ₹80.

Selling price (S.P.) = ₹68.

Discount = M.P. − S.P. = 80 − 68 = ₹12.

$\therefore$ Rate of discount = $\frac{\text{Discount} \times 100}{\text{Marked Price}} = \frac{12 \times 100}{80} = 15\%$

5. (a) Cost price of 200 dozen oranges = 200 × 10 = ₹2000

Transportation cost = ₹500.

$\therefore$ Cost Price (C.P.) = 2000 + 500 = ₹2500.

Selling Price (S.P.) = 200 × 12 × 1 = ₹2400.

Loss = C.P. − S.P. = 2500 − 2400 = ₹100.

$\therefore$ Loss % = $\frac{\text{Loss} \times 100}{\text{C.P.}} = \frac{100 \times 100}{2500} = 4\%$

6. (c) We have, S.P. = ₹10500, gain% = 5%

$\therefore$ C.P. = $\left(\frac{100}{100 + \text{Gain}\%}\right) \times \text{S.P.}$

$= \left(\frac{100}{100 + 5}\right) \times 10500$

$= \frac{100}{105} \times 10500 = ₹10000.$

7. (a) Here, C.P. = ₹1800, loss % = 10%

$\therefore$ S.P. = $\left(\frac{100 - \text{Loss}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 - 10}{100}\right) \times 1800$

$= \frac{90}{100} \times 1800 = ₹1620.$

8. (b) We have,

Cost price (C.P.) = (120 × 80 + 280 + 72 + 120 × 40)

= ₹10000.

Gain % = 8%

$\therefore$ Selling Price (S.P.) of 120 rims

$= \left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 8}{100}\right) \times 10000$

$= \frac{108}{100} \times 10000 = ₹10800.$

Thus, selling price per rim = $\frac{10800}{120} = ₹90.$

9. (c) In the first case, S.P. = ₹31 and loss % = 7%

$\therefore$ C.P. = $\left(\frac{100}{100 - \text{Loss}\%}\right) \times \text{S.P.} = \left(\frac{100}{100 - 7}\right) \times 31$

$= \frac{100}{93} \times 31 = ₹33\frac{1}{3}.$

In the second case, C.P. = $₹33\frac{1}{3}$ and gain % = 5%

$\therefore$ S.P. = $\left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 5}{100}\right) \times \frac{100}{3} = \frac{105}{100} \times \frac{100}{3} = ₹35.$

Alternative Solution

93% C.P of Cricket ball = ₹31

To gain 5%

$\therefore$ 105% of C.P of cricket ball = $\frac{31}{93} \times 105$

= ₹35

10. (a) In the first case, we have, S.P. = ₹35 and gain % = 40%

$\therefore$ C.P. = $\left(\frac{100}{100 + \text{Gain}\%}\right) \times \text{S.P.}$

$= \left(\frac{100}{100 + 40}\right) \times 35$

$= \frac{100}{140} \times 35 = ₹25.$

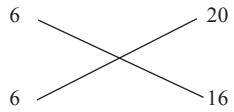
In the second case, C.P. = ₹25 and gain% = 60%

$\therefore$ S.P. = $\left(\frac{100 + \text{Gain}\%}{100}\right) \times \text{C.P.}$

$= \left(\frac{100 + 60}{100}\right) \times 25$

$= \frac{160}{100} \times 25 = ₹40.$

11. (c) Quantity Price



% profit = $\left(\frac{xw}{zy} - 1\right) \times 100\% = \left(\frac{6 \times 16}{4 \times 20} - 1\right) \times 100\%$

$= \frac{16}{80} \times 100\%$

= 20%